



SKILL DUEL PLATFORM

Project Scope & Product Requirements Document

Built with Flutter · Firebase · Flame Engine

VERSION 2.0 — FULL MVP EDITION

v2.0 · April 2026

1. Vision & Concept

Skill Duel is a competitive 1v1 mobile platform where players challenge each other to knowledge and skill battles across their most passionate interests — football, chess, gaming, and more. Think Chess.com's ranked ladder energy, applied to any subject you care deeply about.

Core Insight

SoloLearn proved that async challenge formats are addictive. Chess.com proved ELO ranking systems create obsessive retention. Skill Duel combines both — bringing ranked competitive dueling to communities that have no dedicated battleground: football fans, COD players, chess tacticians.

1.1 The Problem

- Football fans have no competitive knowledge platform — just passive apps and stats pages.
- COD / gaming knowledge is scattered across Reddit and YouTube, never gamified.
- Trivia apps feel like homework. No stakes, no rank, no rival.
- The SoloLearn duel format is addictive but locked inside a coding niche.

1.2 The Solution

- Async 1v1 duels: both players answer independently, winner revealed after both submit.
- ELO rating system: every duel affects your rank. Skill is tracked and meaningful.
- Niche categories: Football, Chess Tactics, COD/FPS, General Gaming.
- Seasons & leaderboards: monthly resets, rank tiers from Bronze to Master.
- Social layer: challenge friends directly or get matched by skill level.
- Daily Challenge: one special free duel per day with a global leaderboard.
- Shareable result cards: viral hooks built into every duel outcome.

2. MVP Scope — What Ships at Launch

MVP Philosophy

The MVP proves the entire core loop end-to-end: register → pick category → duel → result → rank update → share. Every feature below is required for this loop to feel complete and competitive. Nothing has been deferred unless it cannot be built solo in 12 weeks.

2.1 MVP Feature List

Feature	Description	Status
User Authentication	Google Sign-In, email/password, anonymous guest mode via Firebase Auth	MVP
Onboarding Flow	3-step welcome carousel + avatar/username setup after registration	MVP
Category Selection	Football and Chess at launch. COD and Gaming as locked placeholders shown in UI	MVP
Question Bank	300 Football + 200 Chess curated questions. Difficulty tiers: Easy, Medium, Hard	MVP
Direct Challenge	Generate a duel code, share via WhatsApp/Instagram. Opponent joins via code or deep link	MVP
Auto Matchmaking	Match within ± 150 ELO range. Expands to ± 300 after 30s. Open challenge after 60s	MVP
Async Duel Engine	10 questions per duel. 20-second timer per question. Both players answer independently	MVP
Timer Warning State	Red pulsing timer and subtle screen tint when 5 seconds remain	MVP
ELO Rating System	Per-category ELO starting at 1000. K-factor 32 below 1200, 16 above. Cloud Function calculated	MVP
Duel Results Screen	Win and loss states. Score comparison, ELO change, new ELO display	MVP
Answer Review Screen	Post-duel review of all 10 answers with correct/wrong state and explanations	MVP
Share Result Card	Shareable graphic card generated after each duel. One-tap share to WhatsApp/Instagram/X	MVP
Waiting for Opponent	Suspense screen while opponent plays. Score locked until both submit	MVP
Duel Cancelled Screen	Walkover win handling when opponent abandons or never accepts within 24 hours	MVP
User Profile	Avatar, username, ELO per category, win/loss record, recent duels, achievements row	MVP
Global Leaderboard	Top 100 per category. Weekly and All-Time toggle. Podium top 3 display	MVP
Friend System	Add friends by username/link. Challenge friends directly. Online status indicator	MVP

Feature	Description	Status
Duel History	Full paginated history with W/L filter, score, ELO change, category per duel	MVP
Notifications	Push via Firebase Cloud Messaging: challenged, opponent submitted, your turn, daily challenge	MVP
Daily Challenge	One special free duel per day per category. Countdown timer to reset. Global daily leaderboard	MVP
Tutorial Overlay	Coach marks on first duel: timer explanation, answer tap, async explainer. Skip option	MVP
Report Question Modal	Flag bad/wrong questions during or after duel. 5 report reasons + optional note	MVP
Network Error Screen	Offline detection with duel progress saved confirmation and reconnect action	MVP
Upgrade to Pro Screen	Pro subscription pitch: unlimited duels, no ads, advanced stats, exclusive frames	MVP
Purchase Success Screen	Gold celebration screen post-purchase. Pro badge + unlocked benefits checklist	MVP
Settings Screen	Account, notifications toggles, linked accounts, app version, delete account	MVP
Empty State Screen	First-time user empty duel history with single CTA to start first duel	MVP
Dark Mode	Full dark UI, always on. Non-negotiable brand identity	MVP

Deliberately Deferred to V2

Seasons system, rank tier screen, season end summary, custom tournament brackets, community question submissions, chat/trash talk, COD and Gaming categories live question banks, web admin dashboard. These are growth features, not launch requirements.

3. Complete Screen Inventory (36 Screens)

All 36 screens have been designed and have corresponding Stitch prompts. The table below maps every screen to its category, MVP status, and Stitch prompt reference.

Screen	Category	Version
01 — Splash Screen	Auth & Onboarding	MVP
02 — Onboarding Carousel	Auth & Onboarding	MVP
03 — Login Screen	Auth & Onboarding	MVP
04 — Sign Up Screen	Auth & Onboarding	MVP
05 — Avatar Setup	Auth & Onboarding	MVP
06 — Home / Category Select	Core Loop	MVP
07 — Category Detail	Core Loop	MVP
08 — Matchmaking Screen	Core Loop	MVP
09 — Direct Challenge Screen	Core Loop	MVP
10 — Active Question Screen	Core Loop	MVP
11 — Answer Selected State	Core Loop	MVP
12 — Timer Warning State	Core Loop	MVP
13 — Waiting for Opponent	Core Loop	MVP
14 — Duel Cancelled / Walkover	Core Loop	MVP
15 — Win Results Screen	Results	MVP
16 — Loss Results Screen	Results	MVP
17 — Answer Review Screen	Results	MVP
18 — Share Result Card	Results	MVP
19 — Daily Challenge Screen	Daily & Seasons	MVP
20 — Daily Challenge Completed	Daily & Seasons	MVP
21 — Season End Summary	Daily & Seasons	V2
22 — Friends Screen	Social	MVP
23 — Add Friend / Search	Social	MVP
24 — Opponent Public Profile	Social	V2
25 — Notifications Screen	Social	MVP
26 — Duel History Screen	Social	MVP
27 — My Profile Screen	Profile & Progress	MVP
28 — Global Leaderboard	Profile & Progress	MVP
29 — Rank Tier Screen	Profile & Progress	V2
30 — Settings Screen	Utility	MVP
31 — Upgrade to Pro	Utility	MVP

Screen	Category	Version
32 — Purchase Success	Utility	MVP
33 — Report Question Modal	Utility	MVP
34 — Tutorial Overlay	Utility	MVP
35 — Network Error / Offline	Utility	MVP
36 — Empty State	Utility	MVP

4. Technical Architecture

4.1 Full Tech Stack

Technology	Purpose	Priority
Flutter	Cross-platform framework. Single codebase for iOS + Android. Dart language.	Core
Firebase Auth	Google Sign-In, email/password, anonymous auth. First-class Flutter SDK.	Core
Cloud Firestore	User profiles, question bank, leaderboards, duel metadata, friends.	Core
Firebase Realtime DB	Live duel state sync. Low-latency, built for real-time game state.	Core
Firebase Cloud Functions	ELO calculation, duel finalization, receipt verification, anti-cheat.	Core
Firebase Cloud Messaging	Push notifications for challenges, turn alerts, daily challenge reminder.	Core
Firebase Storage	User avatars, generated share result card images.	Core
in_app_purchase (Flutter)	Handles both Google Play Billing and Apple StoreKit from one package.	Core
Flutter Flame	Lightweight 2D engine for timer animations and result screen effects.	Optional
Riverpod	State management. Cleaner async handling than BLoC for this app size.	Core
Hive	Local cache for offline question browsing and profile data.	Core
Dynamic Links / App Links	Deep links for duel codes — opens app directly to duel from WhatsApp.	Core

4.2 Firebase Data Structure

```
Firestore Collections:
users/{uid}
  username, avatar, eloMap: {football: 1050, chess: 980}
  wins, losses, isPro, createdAt, fcmToken

questions/{id}
  category, subcategory, difficulty (1-3)
  question, options[4], answer, explanation
  reportedCount, isActive

leaderboards/{category}/weekly/{uid}
  elo, rank, username, avatar

friends/{uid}/list/{friendUid}
  username, avatar, eloMap, status

Realtime Database:
duels/{duelId}
  status: waiting | in_progress | completed | cancelled
  category, difficulty, createdAt, completedAt
  playerA: { uid, elo, answers[], score, submitted, timePerQuestion[] }
  playerB: { uid, elo, answers[], score, submitted, timePerQuestion[] }
  questions: [q1Id, q2Id, ...q10Id]

matchmaking/{category}/{difficulty}
  queue: [{ uid, elo, timestamp }]

daily/{category}/{date}
  questionIds: [...], playerCount, topScores[10]
```

4.3 App Architecture — Folder Structure

```
lib/
├── core/
│   ├── theme/           # Colors, typography, dark mode tokens
│   ├── utils/           # ELO calculator, share card generator
│   ├── services/        # Firebase wrappers, notification service
│   └── router/          # GoRouter navigation + deep link handling
├── features/
│   ├── auth/            # Login, register, avatar setup screens
│   ├── home/            # Home screen, category cards
│   ├── duel/            # Engine, question UI, timer, matchmaking
│   ├── results/         # Win/loss screen, answer review, share card
│   ├── daily/           # Daily challenge screen + completed state
│   ├── profile/         # My profile, opponent profile, history
│   ├── leaderboard/     # Global + friend rankings
│   └── friends/         # Friends list, add friend, notifications
```

```

├── settings/      # Settings, upgrade to pro, purchase success
├── onboarding/    # Splash, carousel, first-run tutorial
├── models/        # Duel, User, Question, ELO data classes
└── main.dart

```

5. Payment Integration

✓ Key Decision

No custom payment gateway (Stripe, Paystack, etc.) is needed. Google Play and the App Store are the payment processors. You build the pitch UI — the OS handles billing. This saves 3–4 weeks of payment integration work.

5.1 How It Works

Both platforms use the Flutter `in_app_purchase` package — one package handles both Google Play Billing API and Apple StoreKit automatically.

1. User taps 'Upgrade to Pro' on Screen 31 (Upgrade to Pro Screen).
2. Your Flutter code calls the `in_app_purchase` package with product ID `skill_duel_pro_monthly`.
3. The OS presents its native payment sheet — Google Pay or Apple Pay. You do NOT design this screen, the OS owns it entirely.
4. On successful payment, Google/Apple sends a purchase token to your app.
5. Your app sends the token to a Firebase Cloud Function for server-side verification.
6. The Cloud Function calls Google Play Developer API or Apple App Store Server API to verify the receipt.
7. On successful verification, the Cloud Function sets `user.isPro = true` in Firestore.
8. App listens to Firestore in real time, detects `isPro = true`, and navigates to Screen 32 (Purchase Success Screen).

```

// Flutter – trigger purchase
final products = await
InAppPurchase.instance.queryProductDetails({'skill_duel_pro_monthly'});
final purchaseParam = PurchaseParam(productDetails: products.productDetails.first);
InAppPurchase.instance.buyNonConsumable(purchaseParam: purchaseParam);

// Flutter – listen to purchase stream
InAppPurchase.instance.purchaseStream.listen((purchases) {
  for (final purchase in purchases) {
    if (purchase.status == PurchaseStatus.purchased) {
      // Send purchase.verificationData.serverVerificationData to Cloud Function
    }
  }
});

```

```

        verifyReceiptCloudFunction(purchase.verificationData.serverVerificationData);
    }
}
});

// Cloud Function – verify and unlock Pro
exports.verifyPurchase = functions.https.onCall(async (data, context) => {
    const receipt = data.receiptToken;
    const isValid = await verifyWithGooglePlay(receipt); // or Apple
    if (isValid) {
        await admin.firestore().doc('users/' + context.auth.uid).update({ isPro: true });
    }
});
});

```

5.2 Revenue Split

Item	Detail
Google Play Commission	15% on first \$1M revenue per year, 30% above that
Apple App Store Commission	15% for small developers earning under \$1M/year, 30% above
Your net on \$3.99/month	~\$3.39 per subscriber on Google Play (15% tier)
Subscription Product ID	skill_duel_pro_monthly — defined in Play Console and App Store Connect
Free trial	3 days — configured in Play Console / App Store Connect dashboard, no code needed
Restore purchases	in_app_purchase.restorePurchases() — required by both app stores

5.3 App Store Setup Required

- Google Play Console: Create app → Create subscription product skill_duel_pro_monthly → Set price \$3.99 → Enable free trial 3 days → Link to Firebase service account for receipt verification.
- App Store Connect: Create app → Create auto-renewable subscription → Set price tier → Enable 3-day free trial → Download receipt verification private key.
- Both store listings must be live before the in_app_purchase package can return real products during development — use sandbox/test accounts for testing.

6. Core Duel Engine — Deep Dive

6.1 Async Duel Flow

9. Player A selects category + difficulty, taps Start Duel.
10. Matchmaking: either enters friend code or joins auto-match queue.
11. Cloud Function creates duel document in Realtime DB, selects 10 random questions from Firestore matching category and difficulty.
12. Player A sees questions one at a time, 20-second timer each. Answers stored locally then synced.
13. After question 10, Player A sees Waiting for Opponent screen. Score is locked and hidden.
14. Player B receives FCM push notification: 'You've been challenged to a Football duel!'
15. Player B plays same 10 questions. On submission, Cloud Function compares scores.
16. Both players receive FCM push: 'Results are in!' Both see results screen simultaneously.
17. Cloud Function calculates new ELO for both players and writes to Firestore.

6.2 ELO Calculation

```
// Dart – ELO logic (also mirrored in Cloud Function for authority)
double expectedScore(double ratingA, double ratingB) {
  return 1 / (1 + pow(10, (ratingB - ratingA) / 400));
}

double newRating(double rating, double expected, double actual, int k) {
  return (rating + k * (actual - expected)).roundToDouble();
}
// actual: 1.0 = win, 0.5 = draw, 0.0 = loss
// k: 32 if rating < 1200, else 16
// ELO is ALWAYS calculated server-side in Cloud Function – never trust client
```

6.3 Question Bank Schema

```
{
  id: "ftb_001",
  category: "football",           // football | chess | cod | gaming
  subcategory: "premier_league", // granular topic
  difficulty: 2,                  // 1=easy, 2=medium, 3=hard
  question: "Which player has the most PL assists of all time?",
  options: ["Ryan Giggs", "Frank Lampard", "Wayne Rooney", "Cesc Fabregas"],
  answer: "Ryan Giggs",
  explanation: "Ryan Giggs recorded 162 Premier League assists.",
  source: "premierleague.com",    // required for every question
  isActive: true,
  reportedCount: 0,
  createdAt: timestamp
}
```

6.4 Minimum Question Counts at Launch

Category	Count
Football — Premier League	150 questions
Football — World Cup / International	80 questions
Football — African Football / AFCON	70 questions
Chess — Openings & Theory	80 questions
Chess — Famous Games & Players	70 questions
Chess — Tactics & Rules	50 questions
TOTAL AT LAUNCH	500 questions minimum

7. Monetization



Principle

Skill Duel is never pay-to-win. ELO is earned, not bought. Paying users get cosmetics and convenience — never a gameplay advantage. This is non-negotiable for community trust.

Revenue Stream	Description	Version
Skill Duel Pro	\$3.99/month or \$28.99/year. Unlimited duels/day, no ads, advanced stats, exclusive avatar frames, Pro badge on profile.	MVP
3-Day Free Trial	Configured in Play Console / App Store Connect. Removes barrier to first conversion.	MVP
Interstitial Ads	Shown between duels for free users. Capped at 1 per 3 duels. Removed with Pro.	MVP
Restore Purchase	Required by both app stores. <code>in_app_purchase.restorePurchases()</code> call.	MVP
Season Pass	Monthly premium cosmetic track. \$4.99/season.	V2
Cosmetic Packs	Animated result screens, duel backgrounds, rare avatar items. \$0.99–\$2.99.	V2

8. Build Timeline — 14 Weeks to Launch



Updated Timeline

The original 12-week plan has been extended to 14 weeks to account for payment integration, 36-screen UI implementation from Stitch designs, and proper question bank curation. This is a realistic solo developer timeline.

Phase	Duration	Deliverables
Phase 1	Week 1–2	Project setup, Flutter + Firebase config, GoRouter navigation, dark theme system, design tokens from Stitch exports, splash + onboarding screens.
Phase 2	Week 3	Auth flow: Google Sign-In, email/password, anonymous mode, avatar setup, Firebase Auth integration.
Phase 3	Week 4–5	Question bank: Firestore schema, seed 500 questions (Football + Chess), question display UI, timer component, difficulty filtering.
Phase 4	Week 6–7	Duel engine: matchmaking queue, duel session creation, async question flow, answer storage, waiting screen, duel cancellation handling.
Phase 5	Week 8	Results system: win/loss screens, answer review, ELO Cloud Function, share result card generator, Firestore ELO writes.
Phase 6	Week 9	Social features: friends system, add friend, opponent profile, duel history, notifications (FCM setup).
Phase 7	Week 10	Profile, leaderboard, daily challenge with countdown + daily leaderboard, push notification handlers.
Phase 8	Week 11	Payments: in_app_purchase setup, Play Console + App Store Connect config, Cloud Function receipt verification, Pro unlock flow, purchase success screen.
Phase 9	Week 12	Remaining utility screens: settings, report question, tutorial overlay, network error, empty state, upgrade to pro screen.
Phase 10	Week 13	Beta testing with 20–50 real users. Question quality audit. Bug fixes. Performance profiling. Edge case handling.
Phase 11	Week 14	App store submission (iOS + Android). Store listing copy, screenshots, metadata. Launch marketing prep.

9. Launch & Growth Strategy

9.1 Viral Loop



The Share Mechanic

After every duel, both winner and loser get a shareable result card (Screen 18). The card shows score, ELO change, category, and a personal challenge link: 'Can you beat me? skillduel.app/@kofi_duel'. One tap shares to WhatsApp Stories or Instagram Stories. Every share is a free ad. Every viewer is a potential install.

9.2 Launch Channels

- Football: Football Twitter/X and WhatsApp group links. Result cards with tribal team language travel fast.
- Chess: r/chess, Chess.com forums, chess Discord servers. This community loves anything that tests knowledge.
- African market: deep WhatsApp penetration makes the share mechanic extremely effective. Football passion is unmatched.
- Gaming TikTok: one creator showing a COD knowledge duel can produce 50k+ installs overnight.

9.3 Retention Mechanics

- ELO rank: players protect their number obsessively. Losing 30 ELO hurts. That emotion brings them back.
- Daily challenge: one reason to open the app every single day even without an active duel.
- Friend rivalries: @chess_king_99 just hit 1400 ELO — triggers an instant challenge.
- Push notifications: opponent submitted, challenged, daily challenge ready — three daily re-engagement hooks.

10. Success Metrics

Metric	Target & Meaning
D1 Retention	Target >55% — users return the day after install
D7 Retention	Target >35% — strong signal of habit formation
Duels per DAU	Target >3 — users playing multiple duels per session
Challenge Accept Rate	Target >60% of issued challenges accepted within 24h
Daily Challenge Completion	Target >40% of DAU completing the daily challenge
Pro Conversion (30-day)	Target >4% of 30-day active users upgrading to Pro

Metric	Target & Meaning
Share Rate	Target >25% of completed duels generating a share action
Question Report Rate	Target <1% — low rate means question quality is high
Avg Session Length	Target >8 minutes per session
Week 1 MAU → Week 4 MAU Ratio	Target >60% — measures early cohort retention

11. Risks & Mitigations

Risk	Description & Mitigation	Level
Question Exhaustion	Players see repeats. Mitigation: 500 questions at launch, weighted random selection, community submission in V2.	High
Empty Auto-Match Queue	Not enough players for live matching. Mitigation: Direct challenge removes this dependency. Focus on friend invites at launch.	High
Cheating / Googling	20-second timer makes Googling impractical for most questions. App-level detection in V2.	Medium
Bad Question Quality	Destroys trust instantly. Mitigation: Human review of every question, source required, report system ships at launch.	High
App Store Rejection	Standard quiz app, no gambling, no adult content. Low risk.	Low
Payment Receipt Spoofing	Client-side purchase faking. Mitigation: ALL receipt verification in Cloud Functions server-side. Never trust client.	Medium
Push Notification Opt-Out	Users disable notifications, breaking the async loop. Mitigation: In-app notification center as fallback (Notifications screen).	Medium